Jed McKinney

NASA Hubble Fellow at UT Austin

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Summary

Research Charting the role of dust in the growth and evolution of galaxies from the first galaxies through today. Probing dust content and properties to constrain their physical nature, including active galactic nuclei. Using optical, near-infrared, mid-infrared, far-infrared/sub-mm, and radio observations to measure baryonic properties and numerical simulations to build a complete evolutionary picture of galaxy evolution. Support future mission development to build the next generation of observatories.

Mentoring 9 undergrad (6 to grad school), 3 grad, 2 postdocs. 9 student publications. Received Goldwater, NSF GRFP, NSF AAPF

Grants $992k in grant funding secured from NASA, NRAO, and Heising-Simons

Publications 13 first-author, 73 co-author, 8 student-led

Observing 12 PI programs on *JWST*, ALMA, VLA, IRAM, LMT (230 total hours).

Leadership PRIMA Science Affiliate & Galaxy Evolution Working Group Lead, NASA Advisory Council (IRSTIG), Science collaboration leadership roles (COSMOS-Web, choir), Astronomy on Tap host/organizer (Austin, TX)

Education

The University of Massachusetts, Amherst — *Amherst, MA*
Astronomy PhD. Advisor: Alexandra Pope — *Sept 2017 - August 2022*

Tufts University — *Boston, MA*
BS Astrophysics, Math Minor. Advisor: Anna Sajina — *Sept 2013 - August 2017*

Positions

NASA Hubble Fellow — *UT Austin*
Faculty sponsor: Caitlin Casey (2024-2025), Julian Munoz (2025-present) — *Aug 2024 - Present*

Postdoctoral Fellow — *UT Austin*
Faculty sponsor: Caitlin Casey — *Sept 2022 - July 2024*

Caltech/IPAC Visiting Graduate Fellow — *Caltech*
Advisor: Lee Armus — *Sept 2022 - July 2024*

Graduate Research Assistant — *UMass*
Advisor: Alexandra Pope (PhD), Anne Jaskot — *Sept 2017 - July 2022*

Undergraduate Research Assistant — *Tufts*
Advisor: Anna Sajina — *2015 - 2017*

Grants

Heising-Simons Foundation Conference Support funded for $70k — *UT Austin*
New Results Challenging Underlying Assumptions in Galaxy Evolution and Beyond — *2024*
- Funds to host the inaugural conference for a science collaboration that I co-founded.
- Additional $9k from the North American ALMA Science Center for junior participant travel support.

JWST Cycle 3 AR #5213 funded for $326k — *UT Austin*
SCUBADive: A JWST Archival Analysis of the Most Dust-Obscured Galaxies Across Cosmic Time — *2024*
- Archival project to identify every JWST counterpart to bright sub-mm sources in all fields with JWST and sub-mm imaging.
- Two publications, two in-prep. 4 undergraduate projects, 2 current graduate theses, 3+ future graduate theses.

JWST Cycle 2 GO #3224 funded for $554k — *UT Austin*
Measuring Dust Evolution Over the Past 10 Billion Years with $3 - 12$ μm Spectra for 60 High-z Galaxies — *2023*
- The first MIRI LRS survey with JWST
- One published paper (McKinney et al., 2025c), three in-prep. 2 undergraduate projects, 1 graduate thesis, 2+ future graduate theses.

Archival ALMA Support SOSPADA-011 funded for $34k — *UMass*
An archival analysis on the infrared sizes of distant, dust-obscured galaxies — *2021*
- Funded the final year of my dissertation with an archival research grant from the North American ALMA Science Center.

Observing

Pioneering empirical constraint on dust in the early Universe

1 approved PI JWST program: Cycle 2 GO #3224

Spectroscopic diagnostics of cold gas fueling star-formation

4 approved PI ALMA programs: 2021.1.00953.S, 2022.1.00101.S, 2023.1.00462.S, 2024.1.00538.S
2 approved PI VLA programs: 19B-254, 21A-036
1 approved PI IRAM program: 168-22

Dust continuum surveys of sub-millimeter galaxies in JWST legacy extragalactic fields

1 approved PI JWST AR program: Cycle 3 AR #5213
2 approved PI ALMA programs: 2023.1.00170.S, 2025.1.00869.S
1 approved PI LMT program: 2023-S2-US-16

Approved programs that I contributed to as co-investigator:

- 15 ALMA (over 300 hours), 9 JWST (over 300 hours), and several other LMT, GBT, SOFIA, SMA, and VLA programs.

Mentoring

Postdoctoral Fellows

2 former UT grad student mentees who I hired as postdocs before they departed for prize fellowships

- **Olivia Cooper** Former UT Postdoc, now NSF AAPF at CU Boulder. Regular meetings to discuss science and professional development.
- **Lindsay House** Former UT Postdoc, now SkAI Preceptor at UChicago, The SkAI Institute, and Chicago State University.

Graduate Students

Funded by my PI programs, GO #3224 and AR #5213

- **Miriam Eleazer** UMass grad, thesis on GO #3224. Regular meetings to discuss science and career goals.
- **Dhruv Zimmerman** UFlorida grad, visiting grad research fellow at UT under my supervision.
- **Elaine Gammon** (2025-present). Former undergrad mentee. Now externally advised UW NSF GRFP graduate student.

Undergraduate Students

5 to grad school, 7 1st-author publications, 2 Goldwater and 3 NSF GRFP recipients

- **Marcela Camberos Murillo** (2026 - Present), UT Undergrad.
- **Virginia Vanicek** (2024 - Present) UT Undergrad, accepted to UMD with NSF GRFP, advanced ApJ Letter in prep.
- **Elaine Gammon** (2023- 2025). UGA undergrad, UT REU. Undergrad advisee receiving Goldwater and GRFP.
- **Camila Silva** (2024 - 2025). UChicago Undergrad, UT REU. Now UCSC graduate student. Published AAS Research Note.
- **Cassidy Novack** (2025 - 2026) Whitman undergrad, UT Summer RA. In prep RNAAS, Post-bacc at NRAO.
- **Leo Sajkov** Tufts undergrad, worked on GO #3224. 1st author publication, now graduate student at Stanford.
- **Meredith Stone** (2018-2022) 1st author publication as UMass undergrad, Goldwater scholar. Now UArizona grad student.
- **Rucellie Jimenez** (2021-2022) UMass undergrad.
- **Owen Henry** (2019-2020) UMass undergrad, 1st author RNAAS.

Teaching

UT Introductory Astronomy for non-majors — *UT Austin*
Guest lectured extragalactic modules for 100+ undergrads. — *2023*

Summer Pre-College Astronomy — *UMass*
Designed and delivered lectures for rising freshman. — *2018-2019*

Computer Science Workshops for Undergrads — *UMass*
Built astronomy-oriented coding bootcamp for undergrads. — *2018*

Outreach and Service

Astronomy on Tap, Austin TX — *UT Austin*
Host and Organizer — *2023 - Present*

- Host of monthly show with crowds reaching 150+. One of the longest running Astronomy on Tap show in the country.
- Second AoT to ever reach 100 shows.
- Coordinated talks from NASA Headquarters, NASA Astronauts, and astronomers from around the world.

Junior Scientist Working Group Lead — *Virtual*
COSMOS-Web collaboration — *2022 - 2024*

- Led Junior Scientist working group of largest JWST Cycle 1 collaboration. Ran monthly meetings, led creation of code of conduct.

Conference Organizing

UT Austin

Organizer role in 3 conferences

2023 - 2025

- (SOC/LOC, grant coPI) Inaugural `choir` conference on science and mentoring.
- 8th (SOC/LOC) and 9th (SOC) New Horizons Bash Symposium. UT Austin conferences highlighting excellence in postdoctoral research.

Media Appearances

Austin, TX

Public television news, podcasts

2023 - Present

- Telemundo Odessa TV Interview. Television feature on JWST.
- Season 2 Episode 3 of Paradise City Podcast, “Paradise Is... A Starry Night” . Interview on the joy of discovery and community in Astronomy.

Department Roles

UT, UMass

4 committees, including UT Graduate Admissions

2020 - Present

- Several local organizing committees
- Summer REU research mentor (3 years)
- UT Austin 2023 Graduate Admissions Committee
- UMass Astronomy DEI committee chair
- UMass Freshman undergraduate Astronomy advisor
- UMass Graduate student recruitment committee

Membership

Science Collaborations

- choir, Leadership Council and Founding Member (2024 - Present)
- The Cosmic Evolution Survey (COSMOS), COSMOS-Web, COSMOS-3D (2022 - Present)
- The Cosmic Evolution Early Release Science (CEERS) Survey (2022 - Present)
- The Great Observatories All-sky LIRG Survey (GOALS, 2021 - Present)

Professional Organizations

- PRIMA Science Affiliate and Galaxy Evolution Working Group Lead (2025 - Present)
- PRIMA Galaxy Evolution Science Working Group Member (2023 - Present)
- NASA IR Science and Technology Integration Group Leadership Council (2022 - Present)
- New Great Observatories Science Analysis Group (2023 - Present)
- American Astronomical Society (2016 - Present)

Talks

2027	**Invited**, Stars, Black Holes, and Feedback From Cosmic Dawn to Cosmic Noon	*Paris, FR*
2026	**Invited**, Galaxy Formation at its Peak	*Leiden, NL*
2026	**Contributed**, COSMOS-Web 2026 Team Meeting	*CA, USA*
2026	**Invited**, Exploring the Aromatic Universe in the JWST Era	*Ontario, CA*
2026	**Invited**, PRIMA PI Science Overview	*virtual*
2026	**Contributed**, The First Billion Years of the Universe	*CO, USA*
2025	**Contributed**, Dust Universes 2025: the Fifth Pan Dust Conference	*AZ, USA*
2025	**Contributed**, NASA Hubble Fellowship Symposium	*MD, USA*
2025	**Organizer Review Lecture**, New Data that Challenge Underlying Assumptions in Galaxy Evolution	*ME, USA*
2025	**Contributed**, Infrared Fine-Structure Lines	*MN, USA*
2025	**Contributed**, Cosmic Frontiers Center Inaugural Conference	*TX, USA*
2025	**Contributed**, PRIMA Working Meeting	*CA, USA*
2025	**Invited**, UW Galaxy Evolution Seminar	*WA, USA*
2024	**Contributed**, Physical Simulations of PAH Emission in Galaxies II	*VA, USA*
2024	**Contributed**, CFC-CCA Workshop	*TX, USA*
2024	**Invited**, University of Maryland Colloquium	*MD, USA*
2024	**Contributed**, NASA Hubble Fellowship Symposium	*CA, USA*
2024	**Invited**, Cosmic Odysseys	*Crete*
2024	**Contributed**, The Physics and Impact of Astrophysical Dust	*CO, USA*
2024	**Invited**, IA-FORTH Astronomy Seminar	*virtual*
2024	**Contributed**, AAS 243	*LA, USA*
2023	**Contributed**, 10 Years of ALMA: Past, Present, and Future	*Chile*
2023	**Invited**, Texas A&M Astronomy Colloquium	*TX, USA*
2023	**Contributed**, COSMOS Team Meeting	*NY, USA*
2023	**Contributed**, CEERS Team Meeting	*TX, USA*

10 invited, 7 contributed, between 2018-2023

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First-Author Publications

13 first-author publications since 2019

Simulation-Informed Dust Masses in Typical $6 < z < 14$ JWST Galaxies *2026*

McKinney, J., Zimmerman, D., Narayanan, D, et al., 2026, arXiv:2610.37821

JWST Reveals a Candidate Supermassive Black Hole Binary at $z = 4.3$ in the Brightest Sub-millimeter Galaxy in COSMOS-Web *2026*

McKinney, J., Gupta, A., Munoz, J. B., et al., 2026, accepted to ApJL, arXiv: 2609.01711

A JWST MIRI LRS Survey of 37 Massive Star-Forming Galaxies and AGN at Cosmic Noon – Overview and First Results *2026*

McKinney, J., Eleazer, M., and Pope, A., et al., 2026, ApJ, 1001, 93

Modeling Galaxies in the Early Universe with Supernova Dust Attenuation *2025*

McKinney, J., Cooper, O., and Casey, C., et al., 2025, ApJL, 985, 21.

SCUBADive I: JWST+ALMA Analysis of 289 Sub-millimeter Galaxies in COSMOS-Web *2025*

McKinney, J., Casey, C., and Long, A., et al., 2025, ApJ, 979, 229

A Near-Infrared Faint, Far-Infrared-Luminous Dusty Galaxy at $z \sim 5$ in COSMOS-Web *2023*

McKinney, J., Manning, S., and Cooper, O., et al., 2023, ApJ, 956, 72

Broad Emission Lines in Optical Spectra of Hot Dust-obscured Galaxies can Contribute Significantly to JWST/NIRCam Photometry *2023*

McKinney, J., Finnerty, L., and Casey, C., et al., 2023, ApJL, 946, 39

The IR Compactness of Dusty Galaxies Set Star-formation and Dust Properties at $z \sim 0 - 2$ *2023*

McKinney, J., Pope, A., Kirkpatrick, A., et al.,2023, ApJ, 955, 136

Measuring the Total Ultraviolet Light from Galaxy Clusters at $z = 0.5 - 1.6$: The Balance of Obscured and Unobscured Star-Formation *2022*

McKinney, J., Ramakrishnan, V., Lee, K-S., et al., 2022, ApJ, 928, 88

Dust-Enshrouded AGN can Dominate Host-Galaxy-Scale Cold-Dust Emission *2021*

McKinney, J., Hayward, C., and Rosenthal, L., et al., 2021, ApJ, 921, 55

Regulating Star Formation in Nearby Dusty Galaxies: Low Photoelectric Efficiencies in the Most Compact Systems *2021*

McKinney, J., Armus, L., and Pope, A., et al., 2021, ApJ, 908, 238

Measuring the Heating and Cooling of the Interstellar Medium at High Redshift: PAH and [C II] Observations of the Same Star-forming Galaxies at $z \sim 2$ *2020*

McKinney, J., Pope, A., and Armus, L., et al., 2020, ApJ, 892, 119

Neutral Gas and Lyα Escape in Extreme Green Pea Galaxies *2019*

McKinney, J., Jaskot, A., and Oey, S., et al., 2019, ApJ, 874, 52

Co-Author Publications

73 total co-author publications since 2017, 8 student-led publications.

Student Publications

Zimmerman, Dhruv; McKinney, Jed et al., 2026, arXiv:2610.19379 *2026*

Vanicek, Virginia; McKinney, Jed et al., 2026, arXiv:2609.22428 *2026*

Eleazer et al., incl. McKinney, 2026, ApJ, 1004, 67 *2026*

Silva, Camilla; McKinney, Jed et al., 2026, RNAAS, 10, 10 *2026*

Cooper et al., incl. McKinney, 2025, ApJ, 982, 125 *2025*

Sajkov et al., incl. McKinney, 2024, ApJ, 977, 115 *2024*

Stone et al., incl. McKinney, 2022, ApJ, 934, 27 *2022*

Henry et al., incl. McKinney, 2019, RNAAS, 3, 199 *2019*

Significant contribution

Faisst et al., 2026, arXiv:2609.35992 *2026*
Martinez et al., 2026, arXiv:2609.36076 *2026*
Parente et al., 2026, arXiv:2609.35533 *2026*
Lofaro et al., 2026, arXiv:2610.28597 *2026*
Wang et al., 2026, ApJ, 1010, 129 *2026*
Casey et al., 2026, OJAp, 9, 67549 *2026*
Akins et al., 2026, ApJ, 1006, 229 *2026*
Lofaro et al., 2026, A&A, 712, 84 *2026*
Roy et al., 2026, arXiv:2606.23683 *2026*
Narayanan et al., 2026, arXiv:2606.20809 *2026*
Lambrides et al., 2026, NatAs, 10, 868 *2026*
Silverman et al., 2026, MNRAS, 546, 167 *2026*
Franco et al., 2026, ApJ, 999, 200 *2026*
Zavala et al., 2026, ApJ, 998, 36 *2026*
Akins et al., 2026, ApJ, 997, 218 *2026*
Harish et al., 2025, ApJ, 992, 45 *2025*
Liu et al., 2025, MNRAS, 542, 397 *2025*
Akins et al., 2025, ApJ, 991, 37 *2025*
Casey et al., 2025, ApJL, 990, 61 *2025*
Sajina et al., 2025, ApJ, 985, 201 *2025*
Manning et al., 2025, arXiv 2505.09703 *2025*
Gentile et al., 2025, A&A, 697, 46 *2025*
Zavala et al., 2025, Nature Astronomy, 9, 155 *2025*
Gentile et al., 2024, ApJL, 973, 2 *2024*
Franco et al., 2024, ApJ, 973, 23 *2024*
Zavala et al., ApJL, 2024, 977, 9 *2024*
Long et al., 2024, ApJ, 999, 47 *2024*
Long et al., 2024, ApJ, 970, 68 *2024*
Casey et al., 2024, ApJL, 975, 4 *2024*
Mizener et al., 2024, ApJ, 970, 30 *2024*
Casey et al., 2024, ApJ, 965, 98 *2024*
Akins et al., 2023, ApJ, 956, 61 *2023*
Casey & Kartaltepe et al., 2023, ApJ, 954, 31 *2023*
Pope et al., 2023, ApJL, 951, 46 *2023*
Song et al., 2022, ApJ, 940, 52 *2022*
Song et al., 2021, ApJ, 916, 73 *2021*
Alberts et al., 2021, MNRAS, 501, 1970 *2021*
Jaskot et al., 2019, ApJ, 885, 96 *2019*
Bonato et al., 2017, ApJ, 836, 171 *2017*

Minor contribution

Gozaliasl et al., 2026, arXiv:2610.02470 *2026*
Paquereau et al., 2026, arXiv:2606.31978 *2026*
Chworowsky et al., 2026, arXiv:2605.13966 *2026*
Abedini et al., 2026, MNRAS, 548, 337 *2026*
Fox et al., 2026, ApJ, 1002, 162 *2026*
Jecmen et al., 2026, ApJ, 1001, 234 *2026*
Ghaffari et al., 2026, A&A, 708, 129 *2026*
Asada et al., 2026, arXiv:2601.20045 *2026*
Toni et al., 2026, A&A, 707, 87 *2026*
Gozaliasl et al., 2025, A&A, 703, 129 *2025*
Paquereau et al., 2025, A&A, 702, 163 *2025*
Shuntov et al., 2025, A&A, 695, 20 *2025*
Toni et al., 2025, A&A, 697, 197 *2025*
Faisst et al., 2025, ApJ, 980, 204 *2025*
Kalita et al., 2025, MNRAS, 536, 3090 *2025*
Bianchin et al., 2024, ApJ, 965, 103 *2024*

Lambrides et al., 2024, ApJL, 961, 25 *2024*
Kirkpatrick et al., 2023, ApJL, 959, 7 *2023*
Young et al., 2023, ApJL, 958, 5 *2023*
Popescu et al., 2023, ApJ, 958, 12 *2023*
Lai et al., 2023, ApJL, 957, 26 *2023*
Fujimoto et al., 2023, ApJ, 955, 130 *2023*
Rich et al., 2023, ApJL, 944, 50 *2023*
Armus et al., 2023, ApJL, 942, 37 *2023*
Lai et al., 2022, ApJL, 941, 36 *2022*
U et al., 2022, ApJL, 940, 5 *2022*

Selected White Papers

McKinney, Jed and Lieb, Emily, 2026, *High Redshift Dust*, Published in *Summary of Discussion Sessions from The Dusty Universe 2025*. *2026*

McKinney, Jed et al., 2025, *Constraining the Total Infrared Luminosites of the Most Luminous Little Red Dots*. Published in Volume 2 of the PRIMA GO Book. *2025*

Long, A, **McKinney, Jed**, et al., 2025, *Dust in High Redshift Quiescent Galaxies*. Published in Volume 2 of the PRIMA GO Book. *2025*

Delvecchio, I et al., incl. **McKinney, Jed**, 2025, *The AGN nature of Little Red Dots via PRIMA Stacking*. Published in Volume 2 of the PRIMA GO Book. *2025*

Arriscado, L et al., incl. **McKinney, Jed**, 2025, *The metallicity-PAH relationship from the local universe to cosmic noon*. Published in Volume 2 of the PRIMA GO Book. *2025*

McKinney, Jed et al., 2020, *An evolving photoelectric efficiency at cosmic noon?*. Published in Proceedings of the International Astronomical Union, Volume 352, pp. 243-245 *2020*

Jaskot, A and **McKinney, Jed** et al., 2020, *Neutral gas and the escape of ionizing radiation: Lessons from the low-redshift Green Peas*. Published in Proceedings of the International Astronomical Union, Volume 352, pp. 304-308. *2020*